# S.C.O.U.T. — Work Breakdown Structure

Isabella Rodriguez (Electrical Lead) · ECEN 194 · Updated 2026-11-02. Organized by the team's Linear phases and milestones; SCO-### is the Linear issue. Red-edged boxes are my work; grey boxes are teammates' work that my tasks depend on or feed.

Isabella Rodriguez — Electrical | David Chousal Cantu — Firmware / Software | John Ryan Myrdal — Mechanical | Dr. Shoba Krishnan — Advisor

**S.C.O.U.T. coral reef monitoring buoy · Sep 2026 → Jun 2027**

## 1. Subsystem Bring-Up
Phase 1 · Sep 7 – Oct 23

- **1.1** Final component list and housing dimensions ✓ SCO-70
- **1.2** Power budget, block diagram, Rev A schematic ✓
- **1.3** Electronics design review with Dr. Krishnan ✓ SCO-116
- **1.4** Order parts for one Scout and the shore station ✓ SCO-115
- **1.5** Hydrophone part decision ✓ SCO-8

◆ **Phase 1 closed; open bench work moved to Phase 2** Oct 30

## 2. System Integration
Phase 2 · Oct 22 – Dec 6

◆ **Electronics bench bring-up, Stages 1–3** Nov 20

- **2.1.0** Turbidity sensor kit arrived (Mouser) ✓ SCO-189
- **2.1.1** Breadboard bring-up (Tests 1, 3, 11 passed Oct 29) SCO-152
- **2.1.2** Temperature test protocol to advisors (in review) SCO-258
- **2.1.3** Temperature test and lab demo SCO-259
- **2.1.4** Turbidity: current, settle time, inrush, direction SCO-153
- **2.1.5** Turbidity accuracy pass mark and standard SCO-271
- **2.1.6** Internal temp/humidity sensor (SHT40) SCO-60
- **2.1.7** Measure sleep, transmit and receive current SCO-23/154/227
- **2.1.8** Per-sensor power switching (P-FETs) SCO-87
- **2.1.9** Protect the D cell from charging or shorting SCO-150
- **2.1.10** Battery-voltage telemetry SCO-83
- **2.1.11** Pass Stages 1–3; document the bench wiring SCO-157/156

◆ **Power & housing integration** Nov 13

- **2.2.1** Follow-up orders: D cells, diodes, hydrophone, antenna, meter SCO-158
- **2.2.2** Follow-up parts arrive (~1 week shipping)
- **2.2.3** Power budget: 140-byte packet, D-cell model SCO-223/267
- **2.2.4** OTA listen window, acoustic summary decisions SCO-225/226/234
- **2.2.5** Hydrophone amplifier (reuse earlier team design) SCO-216
- **2.2.6** Humidity sensor calibration SCO-204
- **2.2.7** Hydrophone noise-floor and frequency pass marks SCO-272
- **2.2.8** Lid D-cell bay and antenna; housing pressure test SCO-162/177 *(John)*

◆ **Buoy-to-shore link on the bench** Nov 10

- **2.3.1** RFM95 link test: buoy Feather → shore station SCO-155
- **2.3.2** FCC-compliant modem config, bench check SCO-98 *(David)*

◆ **Duty-cycle firmware on the bench** Nov 6

- **2.4.1** 24 h unattended duty-cycle run SCO-160 *(David)*

◆ **Float soak & battery endurance** Dec 2

- **2.5.1** 7 days on two D cells → 12-month extrapolation SCO-159
- **2.5.2** 4-week salt-water float soak (unit 1) SCO-180 *(John)*

◆ **Full system integration (fall goal)** Dec 4

- **2.6.1** End-to-end rehearsal, 48 h: buoy → shore → dashboard SCO-278 *(David)*
- **2.6.2** Integration build: electronics into the buoy SCO-279 *(John)*
- **2.6.3** Integrated MVP test SCO-182 *(All)*
- **2.6.4** MVP hard limit before winter break *(All)*

◆ **Thesis writing & design review** Dec 4

- **2.7.1** Get Dr. Krishnan's bench task list before she leaves SCO-232
- **2.7.2** Present system architecture and demos to advisors SCO-260 *(John)*

## 3. Enclosure & Waterproofing
Phase 3 · Nov 30 – Jan 15

◆ **Corrosion & seal decisions (John)** Dec 11

◆ **Sealed electronics endurance** Dec 18

- **3.2.1** Sealed electronics run, 7–14 days in the housing SCO-161

◆ **Biofouling, coatings & materials (John)** Jan 15

## 4. Field Prototype Deployment
Phase 4 · Jan 18 – Feb 26

◆ **Real-ocean flotation test (John)** Jan 29

◆ **Integrated waterproof, proof-load and tilt (John)** Feb 12

◆ **GPS & tamper detection** Feb 19

- **4.3.1** Add GPS and tamper detection SCO-62

◆ **Range & link reliability** Feb 26

- **4.4.1** Over-saltwater LoRa range, packet delivery SCO-14/21 *(David)*

## 5. Deployment Prep
Phase 5 · Mar 1 – Mar 19

- **5.0** Get the lithium D cells to the site legally SCO-163

◆ **Site & window locked** Mar 5

◆ **Permits & mooring approved** Mar 12

◆ **Hardware shipped to the deployment site** Mar 19

## 6. Live Deployment (site not decided)
Phase 6 · Mar 22 – May 28

◆ **Deployed, link live** Mar 26

- **6.1.1** Electrical health checks: battery, humidity, packets

◆ **Month 1 field check** Apr 23

◆ **Month 2 field check** May 21

◆ **Final report & data archive** May 28

## 7. Thesis & Presentations
Nov 2 – Jun 4

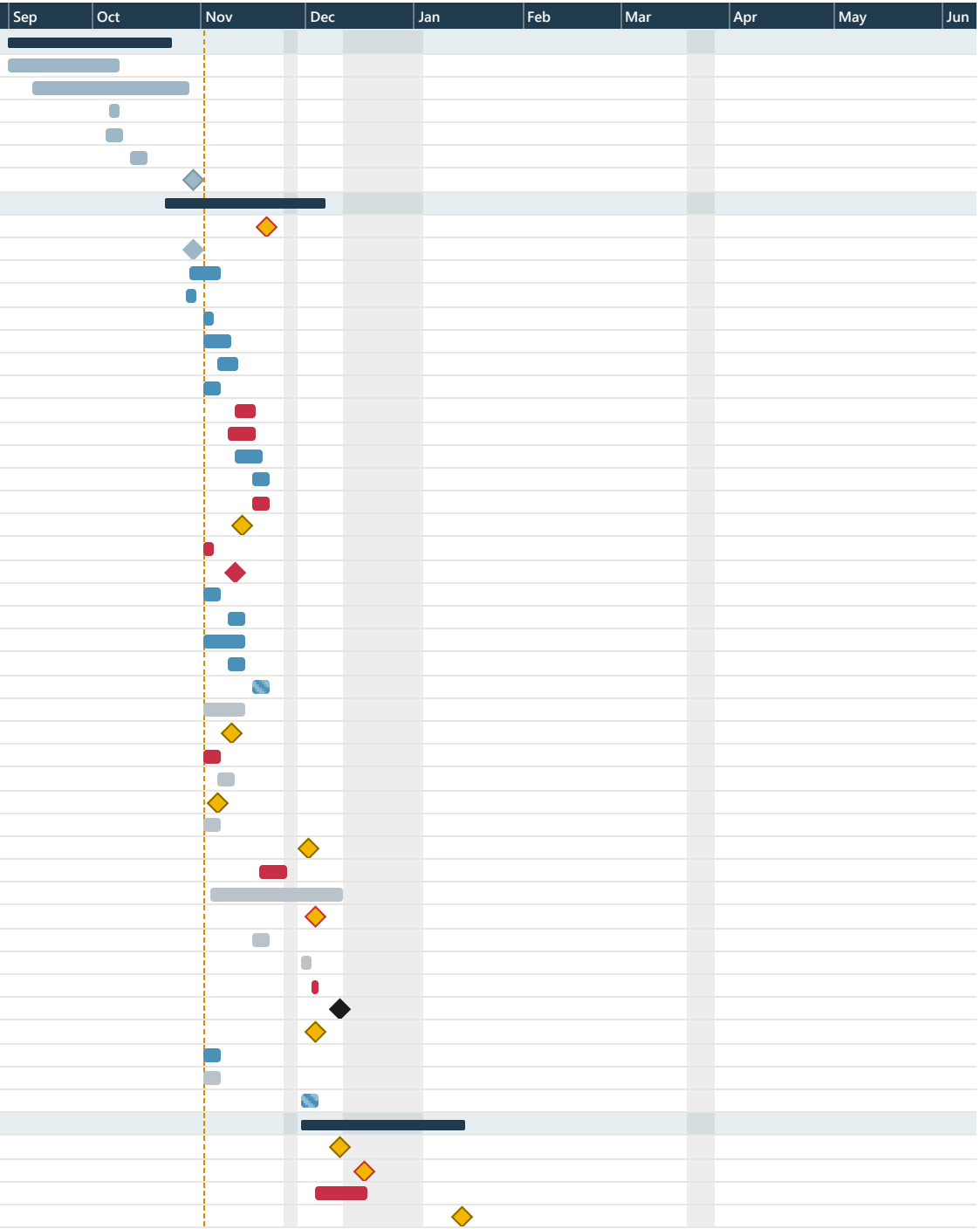
- **7.1** Thesis problem statement, intro, background SCO-206
- **7.2** Advisor meeting: problem, diagram, requirements *(All)*
- **7.3** Ch: Requirements and specifications
- **7.4** Ch: Detailed design — electrical and power
- **7.5** Ch: Electrical testing, results; test-log appendix
- **7.6** Teammates' chapters: John (mech, PM), David (FW, LaTeX) SCO-207/228 *(John, David)*
- **7.7** Senior Design Conference registration closes *(All)*
- **7.8** Discussion, conclusions, abstract, editing *(All)*
- **7.9** Conference slides and poster (my part: electrical) *(All)*
- **7.10** Senior Design Conference *(All)*
- **7.11** Final thesis submitted *(All)*

**2.7.3** Critical design review (CDR)
*(All)*

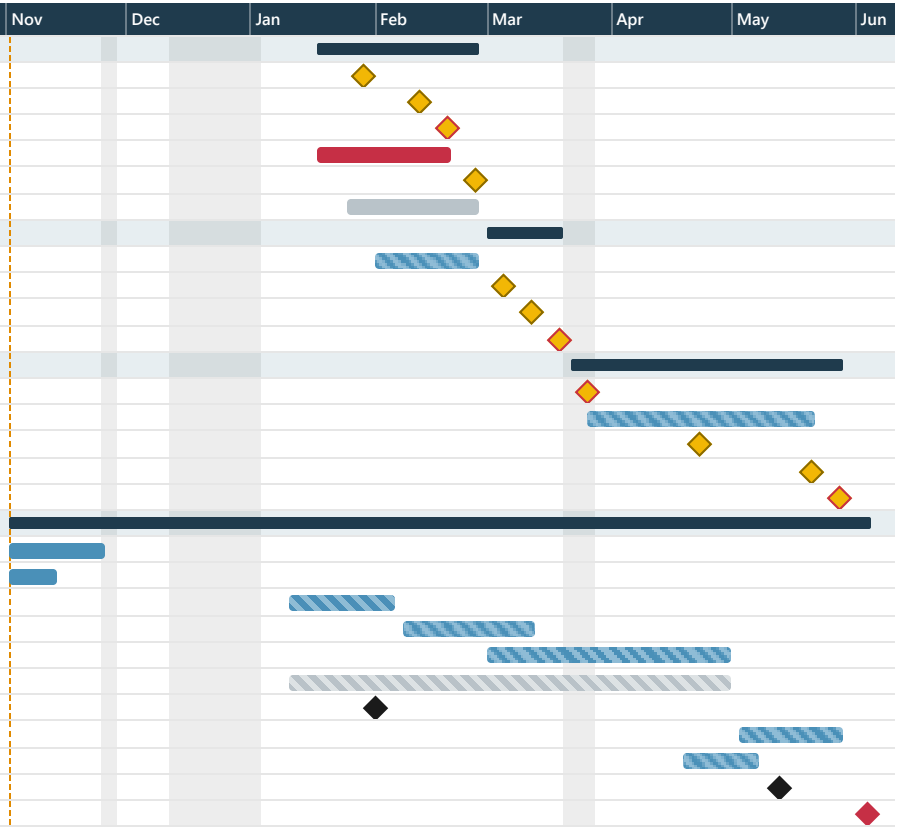
# S.C.O.U.T. — Gantt Chart, Fall 2026 → Spring 2027

Phases and milestone dates match Linear. Several of my Linear due dates from October are past, so this chart shows the realistic dates as of Nov 2 (parts arrival, shipping, Thanksgiving, winter break). "After" = WBS tasks this one waits on.

| WBS | Linear | Task | Owner | After |
|---|---|---|---|---|
| **1** | | **Phase 1 — Subsystem Bring-Up** | | |
| 1.1 | SCO-70 | Final component list and housing dimensions | Isabella | |
| 1.2 | | Power budget, block diagram, Rev A schematic | Isabella | |
| 1.3 | SCO-116 | Electronics design review with Dr. Krishnan | Isabella | 1.1 |
| 1.4 | SCO-115 | Order parts for one Scout and the shore station | Isabella | 1.1 |
| 1.5 | SCO-8 | Hydrophone part decision | Isabella | |
| *1.6* | | ◆ *Phase 1 closed; open bench work moved to Phase 2* | *All* | |
| **2** | | **Phase 2 — System Integration** | | |
| *2.1* | | ◆ *Electronics bench bring-up, Stages 1–3* | | |
| 2.1.0 | SCO-189 | Turbidity sensor kit arrived (Mouser) | Isabella | |
| 2.1.1 | SCO-152 | Breadboard bring-up (Tests 1, 3, 11 passed Oct 29) | Isabella | 1.4 |
| 2.1.2 | SCO-258 | Temperature test protocol to advisors (in review) | Isabella | |
| 2.1.3 | SCO-259 | Temperature test and lab demo | Isabella | 2.1.2 |
| 2.1.4 | SCO-153 | Turbidity: current, settle time, inrush, direction | Isabella | 2.1.0 |
| 2.1.5 | SCO-271 | Turbidity accuracy pass mark and standard | Isabella | 2.1.4 |
| 2.1.6 | SCO-60 | Internal temp/humidity sensor (SHT40) | Isabella | 2.1.1 |
| 2.1.7 | SCO-23/154/227 | Measure sleep, transmit and receive current | Isabella | 2.2.2 |
| 2.1.8 | SCO-87 | Per-sensor power switching (P-FETs) | Isabella | 2.1.4 |
| 2.1.9 | SCO-150 | Protect the D cell from charging or shorting | Isabella | 2.2.2 |
| 2.1.10 | SCO-83 | Battery-voltage telemetry | Isabella | 2.1.9 |
| 2.1.11 | SCO-157/156 | Pass Stages 1–3; document the bench wiring | Isabella | 2.1.7, 2.1.8 |
| *2.2* | | ◆ *Power & housing integration* | | |
| 2.2.1 | SCO-158 | Follow-up orders: D cells, diodes, hydrophone, antenna, m… | Isabella | |
| 2.2.2 | | Follow-up parts arrive (~1 week shipping) | Isabella | 2.2.1 |
| 2.2.3 | SCO-223/267 | Power budget: 140-byte packet, D-cell model | Isabella | |
| 2.2.4 | SCO-225/226/… | OTA listen window, acoustic summary decisions | Isabella | 2.2.3 |
| 2.2.5 | SCO-216 | Hydrophone amplifier (reuse earlier team design) | Isabella | |
| 2.2.6 | SCO-204 | Humidity sensor calibration | Isabella | 2.1.6 |
| 2.2.7 | SCO-272 | Hydrophone noise-floor and frequency pass marks | Isabella | 2.2.5 |
| 2.2.8 | SCO-162/177 | Lid D-cell bay and antenna; housing pressure test | John | |
| *2.3* | | ◆ *Buoy-to-shore link on the bench* | | |
| 2.3.1 | SCO-155 | RFM95 link test: buoy Feather → shore station | Isabella | 2.1.1 |
| 2.3.2 | SCO-98 | FCC-compliant modem config, bench check | David | 2.3.1 |
| *2.4* | | ◆ *Duty-cycle firmware on the bench* | | |
| 2.4.1 | SCO-160 | 24 h unattended duty-cycle run | David | |
| *2.5* | | ◆ *Float soak & battery endurance* | | |
| 2.5.1 | SCO-159 | 7 days on two D cells → 12-month extrapolation | Isabella | 2.2.2, 2.1.11 |
| 2.5.2 | SCO-180 | 4-week salt-water float soak (unit 1) | John | |
| *2.6* | | ◆ *Full system integration (fall goal)* | | |
| 2.6.1 | SCO-278 | End-to-end rehearsal, 48 h: buoy → shore → dashboard | David | 2.3.2, 2.4.1 |
| 2.6.2 | SCO-279 | Integration build: electronics into the buoy | John | 2.1.11, 2.6.1 |
| 2.6.3 | SCO-182 | Integrated MVP test | All | 2.6.2 |
| 2.6.4 | | MVP hard limit before winter break | All | |
| *2.7* | | ◆ *Thesis writing & design review* | | |
| 2.7.1 | SCO-232 | Get Dr. Krishnan's bench task list before she leaves | Isabella | |
| 2.7.2 | SCO-260 | Present system architecture and demos to advisors | John | 2.1.3 |
| 2.7.3 | | Critical design review (CDR) | All | 7.1 |
| **3** | | **Phase 3 — Enclosure & Waterproofing** | | |
| *3.1* | | ◆ *Corrosion & seal decisions (John)* | | |
| *3.2* | | ◆ *Sealed electronics endurance* | | |
| 3.2.1 | SCO-161 | Sealed electronics run, 7–14 days in the housing | Isabella | 2.6.3, 2.5.1 |
| *3.3* | | ◆ *Biofouling, coatings & materials (John)* | | |



| WBS | Linear | Task | Owner | After |
|---|---|---|---|---|
| **4** | | **Phase 4 — Field Prototype Deployment** | | |
| *4.1* | *SCO-184* | ◆ *Real-ocean flotation test (John)* | | |
| *4.2* | *SCO-82* | ◆ *Integrated waterproof, proof-load and tilt (John)* | | |
| *4.3* | | ◆ *GPS & tamper detection* | | |
| 4.3.1 | SCO-62 | Add GPS and tamper detection | Isabella | 3.2.1 |
| *4.4* | | ◆ *Range & link reliability* | | |
| 4.4.1 | SCO-14/21 | Over-saltwater LoRa range, packet delivery | David | 2.3.1 |
| **5** | | **Phase 5 — Deployment Prep** | | |
| 5.0 | SCO-163 | Get the lithium D cells to the site legally | Isabella | |
| *5.1* | | ◆ *Site & window locked* | | |
| *5.2* | | ◆ *Permits & mooring approved* | | |
| *5.3* | | ◆ *Hardware shipped to the deployment site* | | |
| **6** | | **Phase 6 — Live Deployment (site not decided)** | | |
| *6.1* | | ◆ *Deployed, link live* | | |
| 6.1.1 | | Electrical health checks: battery, humidity, packets | Isabella | 6.1 |
| *6.2* | | ◆ *Month 1 field check* | | |
| *6.3* | | ◆ *Month 2 field check* | | |
| *6.4* | | ◆ *Final report & data archive* | | |
| **7** | | **Thesis & Presentations** | | |
| 7.1 | SCO-206 | Thesis problem statement, intro, background | Isabella | |
| 7.2 | | Advisor meeting: problem, diagram, requirements | All | |
| 7.3 | | Ch: Requirements and specifications | Isabella | 7.1 |
| 7.4 | | Ch: Detailed design — electrical and power | Isabella | 7.3 |
| 7.5 | | Ch: Electrical testing, results; test-log appendix | Isabella | 3.2.1 |
| 7.6 | SCO-207/228 | Teammates' chapters: John (mech, PM), David (FW, LaTeX) | John, D… | |
| 7.7 | | Senior Design Conference registration closes | All | |
| 7.8 | | Discussion, conclusions, abstract, editing | All | 7.5 |
| 7.9 | | Conference slides and poster (my part: electrical) | All | 7.5 |
| 7.10 | | Senior Design Conference | All | 7.9 |
| 7.11 | | Final thesis submitted | All | 7.8, 6.4 |




Linear phase · Linear milestone · My task · My critical path · Teammate's task · Estimated · Done · Breaks · Today

**Critical path (my work):** 2.2.1 follow-up orders (overdue in Linear, placing this week) → parts arrive ~Nov 11 → 2.1.7–2.1.8 current measurements and sensor switching → 2.1.11 pass bench Stages 1–3 (Linear said Nov 6; realistic Nov 20) → John's integration build after Thanksgiving → **2.6.3 MVP Dec 4** (hard limit Dec 11) → 3.2.1 sealed endurance run, done Dec 18 → 4.3.1 GPS and tamper detection → live deployment Mar 26 → final report May 28 → final thesis. The 7-day battery run (2.5.1) runs in parallel and must finish by Nov 25. The live deployment site is not decided yet.

# S.C.O.U.T. — Accountability Matrix (RACI)

Team roles for the whole project, grouped by the Linear phases. One A (Accountable) per row.

| Task | Linear phase | Isabella Electrical | David Firmware/SW | John Mechanical | Dr. Krishnan Advisor |
|---|---|---|---|---|---|
| Project planning, Linear upkeep, weekly reports | All | R | R | AR | I |
| Budget, parts orders and purchase log | All | R | R | R | A |
| Safety (hazard form, D-cell protection) | 1–2 | R | I | R | A |
| Electrical design: power budget, schematic, block diagram | 1–2 | AR | C | C | C |
| Electronics bench bring-up and power testing (Stages 1–3) | 2 | AR | C | I | C |
| Buoy-to-shore radio link | 2, 4 | AR | R | I | I |
| Firmware, shore station and dashboard | 2–4 | C | AR | I | I |
| Buoy structure, housing and waterproofing | 2–3 | C | I | AR | I |
| Battery and sealed electronics endurance | 2–3 | AR | C | C | I |
| System integration and MVP test | 2 | R | R | AR | C |
| Field prototype tests, deployment prep and live deployment | 4–6 | R | R | AR | C |
| Thesis (each writes their own subsystem chapters) | 2–6 | R | R | R | A |
| Design reviews and Senior Design Conference | 2–6 | R | R | AR | C |

**R — Responsible**
Does the work. At least 1 per task.

**A — Accountable**
Final review before it's complete. Only 1 per task.

**C — Consulted**
Gives input from their area.

**I — Informed**
Kept up to date on progress.